

Rodrigo da Motta Cabral de Carvalho

📍 Currently in São Paulo, SP, Brazil ✉️ rodrigodamottacc@gmail.com ☎️ +55 21 987997890 in rodrigo-da-motta-c-de-carvalho
🔗 Rodrigo-Motta 📄 Personal Page

Education

- MS - Neuroscience and Cognition, Federal University of ABC** [🔗](#) Grande São Paulo, SP, Brazil
Sep 2023 – July 2025
- **Grade:** A/A;
 - **Thesis:** A Graph Neural Network Approach to Investigate Brain Critical States.
- MS Visiting Research Student - Biomedical Engineering & Imaging Sciences, King's College London** [🔗](#) Greater London, UK
Jul 2024 – Mar 2025
- Awarded a prestigious scholarship to work at the [MeTrICS Lab](#) [🔗](#) (Machine Learning for Translational Imaging and Cortical Surfaces).
- BS - Physics, University of São Paulo** [🔗](#) São Paulo, SP, Brazil
Jan 2019 – Dec 2022
- Best university in South America. [🔗](#)
 - **Grade:** 8.6/10 (Top 15%)
 - **Extracurricular:** Community-Building in Data Science at Hackerspace.IFUSP

Experience

- Research Scientist, CloudWalk Inc.** [🔗](#) São Paulo, SP, BR
April 2025 – Ongoing
- CloudWalk is an AI Fintech unicorn where I:
- Pre-trained finance foundation models using custom transaction-event masking, establishing evaluation, activation steering, and downstream validation pipelines for production fraud and credit-scoring models.
 - Built task-specific evaluation suites and agent benchmarks to test autonomous decision-making and self-healing reliability for a vending machine, achieving 8+ months of near-zero human-in-the-loop operation and improving in our vending machine benchmarks (evaluating multiple dimensions from buying products, marketing, and sales).
 - Led a project that fine-tune LLMs with data from altered states of consciousness to explore shifts in internal representations. Using information theory and embedding dynamics to draw connections between machine and human cognition. Developed new psychometric evaluation methods for LLMs and AI agents, evaluating more specific measures than classical benchmarks.
 - Developed representation-level evaluation methodologies using information theory and embedding dynamics to assess model robustness, safety steering, and out-of-distribution generalization. The model is currently on production and deployed to +1M users.
- Research Intern, King's College London** [🔗](#) Greater London, UK
Jul 2024 – March 2025
- Advanced AI applications in psychiatry at MeTrICS Lab by applying Deep Learning models (e.g., Spatial-Temporal Graph Neural Networks) on GPU clusters to classify psychiatric disorders using large-scale fMRI data from a Brazilian cohort. Achieved high diagnostic accuracy and identified key brain network contributions relating to psychiatric scores.
- **Advisor:** Dr. Emma Claire Robinson – King's College London (KCL)
 - **Co-Advisor:** Dr. Walter Pinaya - Hologen AI & KCL
- Graduate Research Scholar, FAPESP** [🔗](#) São Paulo, SP, BR
Sep 2023 – March 2025
- Applied LLMs for semantic analysis in psychology, focused on semantic trajectory over neurodegenerative diseases and time perception. Also developed Graph Neural Networks with complex systems simulations to model brain dynamics under acute and subacute effects of ayahuasca. Managed the project in an international team in a open-source project.
- **Advisor:** Prof. Dr. João Ricardo Sato - Federal University of ABC (UFABC)
 - **Co-Advisor:** Dr. Walter Pinaya - Hologen AI & KCL
- Data Scientist, Artefact** [🔗](#) São Paulo, SP, BR
Jul 2022 – Sept 2023
- Enabled business intelligence for the North American tire market by developing and deploying executive-level forecasting models for all tire SKUs and estimated variables impact, such as inflation. Managed the end-to-end data pipeline—from accessing raw data to final model deployment. Additionally, led community-building initiatives by designing a Data Science meetup and the global journal clubs.
- Undergraduate Researcher, Brazilian Institute of Neuroscience and Neurotechnology (BRAINN)** [🔗](#) Campinas, SP, BR
Jul 2021 – Jul 2022
- Employed complex system models to better understand brain rest state hemodynamics with functional near-infrared spectroscopy. Developed open-source tools and methods.
- **Advisor:** Prof. Dr. Rickson Coelho Mesquita - UNICAMP & University of Birmingham

Publications

Toro-Hernandez, F., Filho J., **Cabral-Carvalho, R.** (2026). Characterizing Human Semantic Navigation in Concept Production as Trajectories in Embedding Space. The Fourteenth International Conference on Learning Representations (ICLR) 2026, <https://openreview.net/forum?id=QQVmlR97sf> 2026

Cabral-Carvalho, R., Pinaya, W. H. L., Sato, J. R. (2025). A Graph Neural Network Approach to Investigate Brain Critical States Over Neurodevelopment. *Network Neuroscience* (Cambridge, Mass.), 9(2), 761–776. https://doi.org/10.1162/netn_a_00451 2025

Falconi-Souto, A., **Cabral-Carvalho, R.M.**, Fujita, A. et al. (2025). Inferences on the Watts-Strogatz Model: A Study on Brain Functional Connectivity. *Neuroinform* 23, 57. <https://doi.org/10.1007/s12021-025-09756-z> 2025

Paris-Colombo, G., **Cabral-Carvalho, R.M.**, Toro-Hernández, F.D. (2026). Comparing Semantic Navigation in Humans and Large Language Models using Natural Language Processing. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48. [arXiv:2607.12195](https://arxiv.org/abs/2607.12195) 2026

Bonifácio, T. A. S., **Cabral-Carvalho, R. M.**, & Cravo, A. M. (2026). Self-report measures of subjective time: An overview of existing measures and their semantic similarities. *Psychonomic Bulletin & Review*, 33(218). 2026

Pre-prints

Leis, B., **Cabral-Carvalho, R. M.** (2026). RCWT: Measuring Task-Budget Displacement from Coordination Content in LLM Calls. *arXiv preprint arXiv:2607.12216*. <https://doi.org/10.48550/arXiv.2607.12216> 2026

Cabral-Carvalho, R. M., Palhano-Fontes, F., Araujo, D. B., Sato, J. R. (2025). Ayahuasca shifts brain dynamics toward higher entropy: Persistent elevation of Ising temperature correlates with acute subjective effects. *bioRxiv*. <https://doi.org/10.1101/2025.04.25.650509v1> 2025
(Under review at *Network Neuroscience*).

Conferences, Talks and funding

Scholarships & Funding

- Undergraduate Researcher (FAPESP Scholarship) [↗](#),
- Graduate Researcher (FAPESP Scholarship) [↗](#),
- KCL Research Internship Abroad (BEPE Scholarship): 20k USD [↗](#).
- CloudWalk AI Research Grant: 6k USD [↗](#)

Conferences:

- Publication and Poster at **ICLR 2026** (Rio de Janeiro, Brazil) [↗](#)
- Poster at **Stanford Graph Learning Workshop**: AgentMarket: Generative Agent in Dynamic Market Networks (Palo Alto, USA) [↗](#)
- Talk at **Brazilian Society of Neuroscience and Behaviour 2025**: Quantifying Subjective Experiences with LLMs (São Paulo, Brazil)
- Talk at **Brain Modes 2025** (Toronto, Canada) [↗](#)
- Poster at **Brain Modes 2024** (Bilbao, Spain)
- Poster at **Organization for Human Brain Mapping (OHBM) 2024** (Seoul, South Korea)
- Oral presentation at **10th Brazilian Institute for Neuroscience and Neurotechnology (BRAINN) Congress 2024** (UNICAMP, Brazil)

Awards

- Best Oral Presentation, 10th BRAINN Congress 2024 (UNICAMP, Brazil)

Invited Talks (on site):

- **Yonsei University, South Korea** (Prof. Byung-Hoon Kim, [NAIPL](#) [↗](#))
- **Imperial College London** (Dr. Pedro Mediano)
- **University of Oxford** (Prof. Rui Costa & [NeuroAI group](#) [↗](#))
- **University of Zurich** (Prof. Susanne Wegener & Prof. Nicolas Langer)

Extra Activities

Content Creator (blog posts)	Jan 2023 - Ongoing
• As a writer, I am inspired to communicate and spark curiosity about Neuroscience, Cognition, Physics, Data Science and AI. Content ↗	
Data Science Coordinator Hackerspace.IF-USP ↗	2020 - Ongoing
Organize free seminars and tech courses at USP, including a week-long Deep Learning intensive featuring 5 expert speakers for 60 participants. [Course Content] ↗	

Languages

English:	IELTS 7.5 (Academic)	Portuguese:	Native	Spanish:	Basic (No proficiency)
-----------------	----------------------	--------------------	--------	-----------------	------------------------